Teacher answers and topic guidance

Accept scientifically equivalent answers in the learner's usual mode. Award each distinct point once. Keep answers hidden during independent attempts.

Starter

1. Active transport.
2. Water.
3. The nucleus.

Full and faded sequence

Interphase; prophase; metaphase; anaphase; telophase; cytokinesis. The faded step is anaphase: the two copies of each chromosome are pulled to opposite ends. DNA is copied before mitosis, in interphase.

Order and justify

Interphase, prophase, metaphase, anaphase, telophase, cytokinesis. The four middle phases form mitosis. Copy first: the copies must exist before separation. Separate: one copy of each chromosome goes to each new nucleus. Split last: the final split produces two separate new body cells. The builder previews each chosen stage, checks the first out-of-place position and checks the selected scientific reason. Reveal order replaces the arrangement with the scientific order; Reset restores the starting mixed sequence and clears the reason.

Two faulty models

Fault 1: copy DNA in interphase before mitosis. Fault 2: in anaphase send one copy of every chromosome to each end. A later correct cell split cannot repair missing or unevenly separated chromosome copies.

Six-chromosome hinge

C: six in each daughter, with the same set as the original body cell. A, three: confuses this ordinary body-cell model with chromosome halving. B, twelve: treats copied material as the number retained by each daughter. D, none: leaves the copies in one cell instead of distributing them.

About three in four secure responses: move on. Otherwise re-model with two blue and two orange paper strips, labelled A and B, then use the parallel case. A starting set of eight gives eight in each daughter.

Transfer to repair

DNA is copied before matching copies separate into two nuclei. After the cell splits, matching new body cells can replace damaged cells. Accept the same explanation through a model demonstration.

Science scope and misconceptions

Mitosis separates copied chromosomes into two nuclei. Cytokinesis splits the cell. The model concerns ordinary body-cell division. It does not teach the full molecular detail, asexual reproduction or chromosome halving. Do not describe mitosis alone as the whole final split.

Copy before separation: both new nuclei need a matching set. Uneven separation can leave one nucleus with an extra copy and another lacking it. New matching body cells support growth and replacement of damaged cells.

Arrival Supported

1. Two cells.
2. The one cell on the left.
3. The inner outline labelled nucleus.
4. One cell has become two; each new cell has a matching labelled set.

Arrival Independent

1. Two.
2. The nucleus.
3. Both cells have A and B: the sets match.
4. One cell became two, each with a matching chromosome set.

Arrival Aspire

1. Two separate cells.
2. The nucleus.
3. Each new cell has both types, A and B, so the sets match.
4. It shows one cell becoming two with matching sets; it does not show the full molecular detail or every real chromosome.

Exit Supported

1. So matching copies exist for both new nuclei.
2. They can replace damaged cells during repair.

Exit Independent

1. DNA is copied before mitosis, and one copy of every chromosome separates into each new nucleus before the cell splits.
2. More new body cells support growth; new body cells also replace damaged cells during repair.

Exit Aspire

1. No. Splitting the cell does not redistribute chromosome copies; one new nucleus can gain a copy while another lacks it.
2. Copy DNA in interphase before mitosis. The copies must exist before they can separate into matching sets.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, mitosis and cytokinesis. Say why copying must come first. [3]

Interphase before mitosis; cytokinesis last; copying first supplies copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete: DNA is ___; matching copies ___; each new nucleus has a matching ___. [3]

DNA is copied; matching copies separate; each new nucleus has a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Damaged skin needs new body cells. Use copy, separate and split to explain how new matching cells help. [3]

Copy DNA before matching copies separate into two nuclei; split the cell to form matching new body cells; these can replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order the six named stages. Mark the four mitosis phases and justify why interphase comes first. [4]

Interphase first and cytokinesis last; prophase, metaphase, anaphase, telophase in order; those four identified as mitosis; DNA copying in interphase supplies matching copies before separation.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how DNA copying and chromosome separation give each daughter cell a matching set. [3]

DNA is copied before mitosis; one copy of each chromosome goes to each new nucleus; after the final split both daughter cells have matching sets.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Explain how this cell-cycle sequence supports growth and the repair of damaged skin. [3]

Copying followed by even separation and cell division produces matching new body cells; more body cells support growth; new body cells can replace damaged cells during repair.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Predict what happens if copied chromosomes separate unevenly. Explain why a correct final split cannot fix the sets. [3]

One new nucleus may gain a copy while another lacks it; the sets no longer match; the final split does not redistribute the chromosomes between nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why copying DNA after mitosis would break the sequence needed for matching new body cells. [3]

DNA copying must precede mitosis; each new nucleus needs one copy of every chromosome; copying after separation cannot supply the missing matching copies during that earlier separation.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Compare the two faulty models: copying skipped, and both A copies sent to one end. Repair each first error, then link the repaired sequence to new cells for repair. [4]

Copy DNA in interphase before mitosis; in anaphase send one copy of each chromosome type to each end; the later final split produces matching new body cells; these can replace damaged cells.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose the order of interphase, mitosis and cytokinesis. Give one reason why copying comes first. [3]

Interphase before mitosis; cytokinesis last; DNA copying first supplies copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Each new body cell needs A and B. Choose one copy of each or both A copies at one end. Explain your choice. [2]

One copy of each type to each end; each new nucleus then has a matching A and B set.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose new matching cells or uneven chromosome sets to help replace damaged skin cells. Use copy, separate and split to show why. [3]

New matching cells; DNA is copied and the copies separate into matching sets before the final split; new body cells can replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt